# EART11300

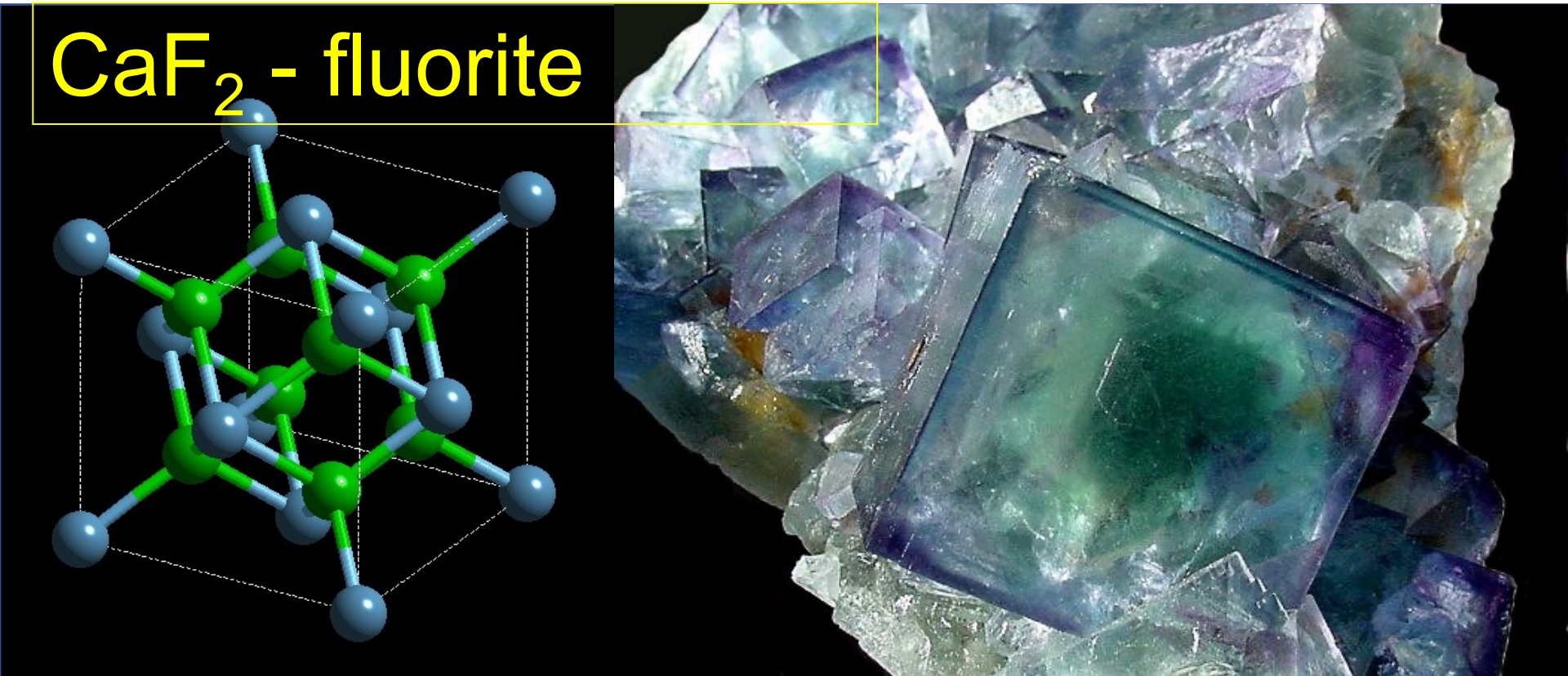
## Physical properties of minerals

## Video 1: What is a mineral?

Video 2: crystal **habit**.
Video 3: **cleavage** in minerals.
Video 4: **colour** and **hardness** of minerals.

# What is a mineral?

Naturally-occurring, inorganic solids, with characteristic chemical compositions (either fixed, or varying within specific limits), and **crystalline structure**.



Euhedral garnet (almandine) crystals
- museum-quality specimens
//www.minerals.net/







Gem-quality diopside crystals, showing prismatic habit and pale green colour

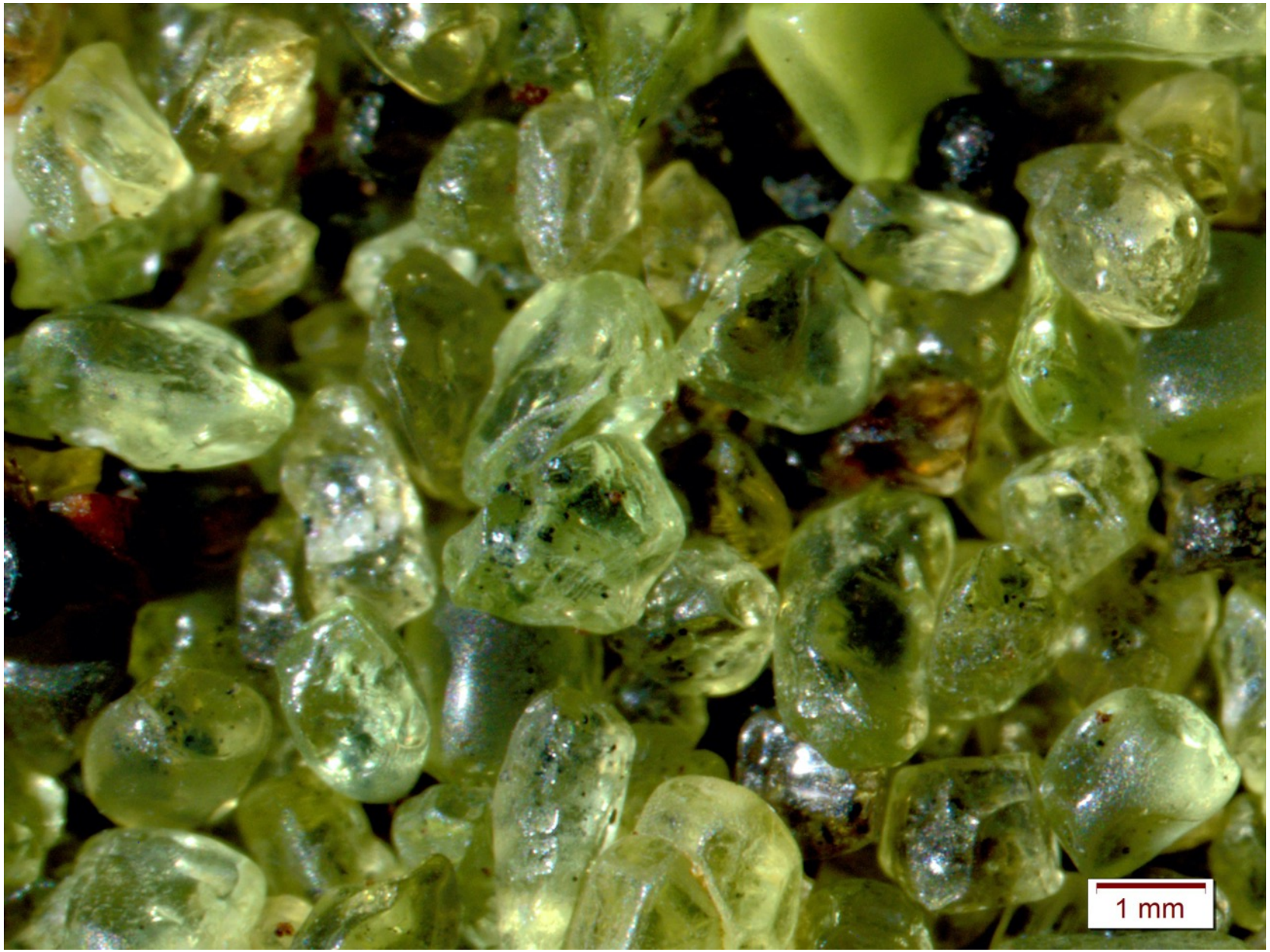
//www.marinmineral.com/





Polished gem-quality olivine crystals, showing characteristic colour

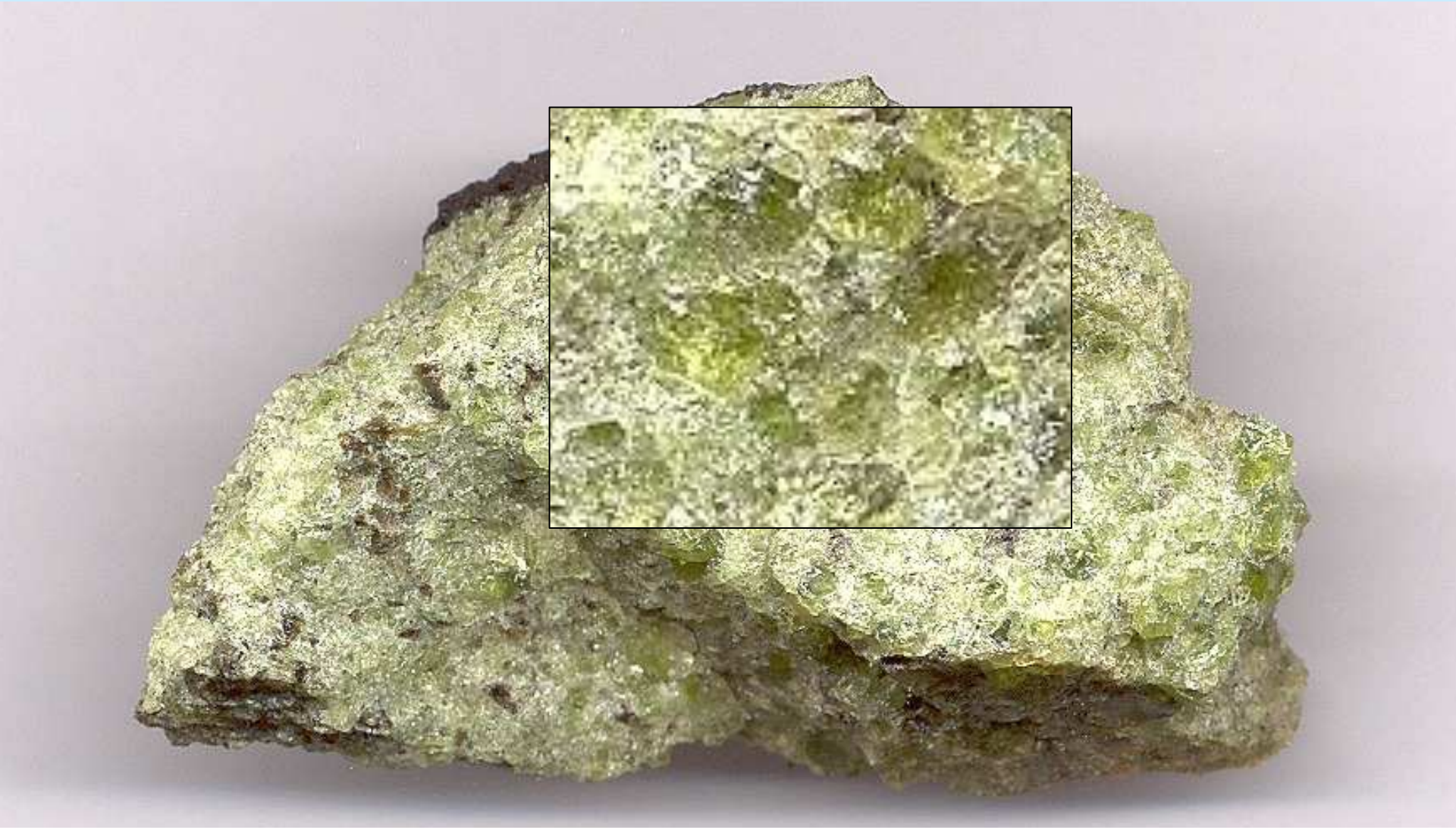



1 mm

Olive-green olivine (forsterite) in a peridotite



Olive-green olivine (forsterite) in a peridotite



Euhedral almandine porphyroblasts in a chlorite schist, showing dodecahedral form



Almandine porphyroblasts in an altered eclogite, with rims of retrogressive hornblende



# What else is in the definition of a mineral?

1. Naturally-occurring
2. Inorganic
3. Solid
4. With characteristic chemical composition (either fixed, or varying within specific limits)

# That was the definition of a mineral. Now what is a **rock**?

- Naturally-occurring solid mass of minerals.
- May be made of a single mineral or (more commonly) >1 mineral.

Quartzite is made of a mass of quartz crystals



Granite is made of quartz + other minerals



Here are a couple of questions for you to think about (answers on next page):

1. Is ice ($H_2O$) a mineral?
2. Is volcanic glass a mineral or a rock (or neither)?

# Is ice a mineral?

Yes, it occurs in nature, it is solid, it is inorganic (it does not contain carbon), and it has a crystalline structure (if it didn't, it wouldn't form beautifully symmetrical snowflakes).



# Is volcanic glass a mineral or a rock (or neither)?

It is not a mineral, as it does not have a crystalline structure. Glass is essentially super-cooled liquid, and has an irregular atomic structure.

As rocks are made of minerals, it is technically not a rock either, but rocks are allowed to contain glass, as long as it is all naturally-occurring.



'Pele's tears' from Hawaii